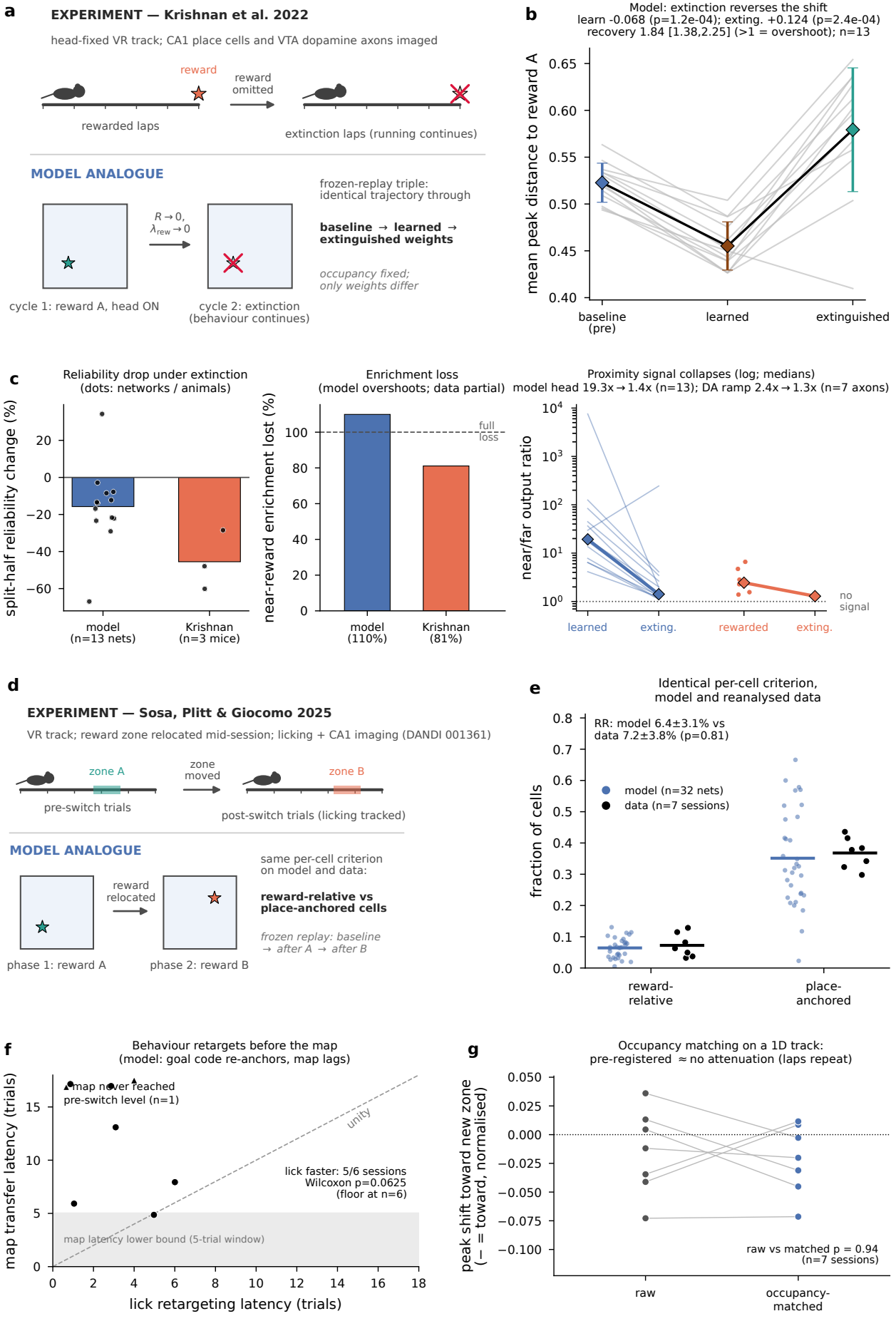

a
EXPERIMENT — Krishnan et al. 2022
head-fixed VR track; CA1 place cells and VTA dopamine axons imaged
reward
reward omitted
rewarded laps
extinction laps (running continues)
MODEL ANALOGUE
frozen-replay triple: identical trajectory through
baseline → learned → extinguished weights
occupancy fixed; only weights differ
R → 0, λrew → 0
cycle 1: reward A, head ON
cycle 2: extinction (behaviour continues)
b
Model: extinction reverses the shift
learn -0.068 (p=1.2e-04); exting. +0.124 (p=2.4e-04)
recovery 1.84 [1.38,2.25] (>1 = overshoot); n=13
mean peak distance to reward A
baseline (pre)
learned
extinguished
c
Reliability drop under extinction (dots: networks / animals)
split-half reliability change (%)
model (n=13 nets)
Krishnan (n=3 mice)
Enrichment loss (model overshoots; data partial)
near-reward enrichment lost (%)
full loss
model (110%)
Krishnan (81%)
Proximity signal collapses (log; medians)
model head 19.3x → 1.4x (n=13); DA ramp 2.4x → 1.3x (n=7 axons)
near/far output ratio
no signal
learned
exting.
rewarded
exting.
d
EXPERIMENT — Sosa, Plitt & Giocomo 2025
VR track; reward zone relocated mid-session; licking + CA1 imaging (DANDI 001361)
zone A
zone moved
zone B
pre-switch trials
post-switch trials (licking tracked)
MODEL ANALOGUE
reward relocated
phase 1: reward A
phase 2: reward B
same per-cell criterion on model and data:
reward-relative vs place-anchored cells
frozen replay: baseline → after A → after B
e
Identical per-cell criterion, model and reanalysed data
RR: model 6.4±3.1% vs data 7.2±3.8% (p=0.81)
model (n=32 nets)
data (n=7 sessions)
fraction of cells
reward-relative
place-anchored
f
Behaviour retargets before the map (model: goal code re-anchors, map lags)
map never reached pre-switch level (n=1)
unity
lick faster: 5/6 sessions
Wilcoxon p=0.0625
(floor at n=6)
map latency lower bound (5-trial window)
map transfer latency (trials)
lick retargeting latency (trials)
g
Occupancy matching on a 1D track:
pre-registered ≈ no attenuation (laps repeat)
peak shift toward new zone (− = toward, normalised)
raw vs matched p = 0.94
(n=7 sessions)
raw
occupancy-matched